

AQUABC Deep Process Rate Analysis — Script Documentation

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1. Overview

The AQUABC Deep Process Rate Analysis toolkit is a suite of Python scripts that provides comprehensive post-simulation quality assurance for the ESTAS-AQUABC ecological water quality model. It reads model output files (state variables and process rates), performs 16 systematic checks, and generates structured text reports and publication-quality PDF documents.

The toolkit validates numerical correctness (NaN/Inf, sign violations, Euler integration), biogeochemical consistency (stoichiometric ratios, mass-balance closure, cross-variable coupling), and ecological plausibility (seasonal patterns, limitation factors, spatial coherence).

2. File Architecture

Script Name	Category	Description
aquabc_analysis_utils.py	Shared utilities	Constants (state variable names, box IDs, stoichiometric ratios), data loading functions (load_process_rates, load_state_vars), helper functions (get_slot_col, compute_kinetic_deriv, compute_numerical_derivative). Imported by all other scripts.
process_rate_slot_map.py	Slot mapping	Complete mapping of all 1,080 process rate slots to their Fortran descriptions, derivative formulas, and sign conventions. Extracted from aquabc_II_pelagic_model.f90. Contains SLOT_MAP dict (36 state variables × 30 slots) and DERIVATIVE_SIGNS dict.
deep_process_rate_analysis.py	Main analysis	16-check analysis engine. Contains check_1 through check_16 functions plus check_cross_box_spatial. run_analysis() orchestrates all checks across all boxes. print_summary() generates structured text output. Can run standalone via CLI.
deep_state_vs_process_crosscheck.py	Cross-check script	10-check independent crosscheck focusing on state-vs-process consistency. Overlaps with checks 9-16 of the main analysis but uses different thresholds and detection algorithms for independent verification.
generate_deep_pr_pdf.py	PDF generator (main)	Generates comprehensive PDF report from the 16-check analysis. Includes executive summary, per-check detailed tables, findings, and recommendations. Uses reportlab.
generate_crosscheck_pdf.py	PDF generator (crosscheck)	Generates PDF report for the 10-check crosscheck analysis.
generate_analysis_results_pdf.py	PDF generator (results)	Generates focused results report documenting what was found, what was fixed, and what remains. Emphasizes interpretation over raw data.

Dependency graph:

All scripts import from aquabc_analysis_utils and process_rate_slot_map. PDF generators import from their respective analysis scripts. No circular dependencies exist.

3. Data Flow

Input files (per box, in OUTPUTS/ directory):

PELAGIC_BOX_XXXXX.out	State variables	Header line + N rows × 37 columns (TIME + 36 state vars). Daily output.
PELAGIC_BOX_XXXXX_PROCESS_RATES.out	Process rates	N rows × 1081 columns (TIME + 36 vars × 30 slots = 1080). Daily output.

Processing pipeline:

1	Load	numpy.loadtxt reads state and process rate files for each box
2	Validate	Check for NaN, Inf, unexpected dimensions
3	Compute derivatives	Kinetic dC/dt from signed process rate sums; numerical dC/dt from concentration forward differences
4	Run 16 checks	Each check returns a list of finding dicts with severity, variable name, and diagnostic values
5	Aggregate	Collect per-box results; run cross-box spatial comparison
6	Report	Print structured text summary; optionally generate PDF

4. Detailed Check Descriptions

Check 1: Rate Statistics

`check_1_rate_statistics(time, rates, box_id)`

Computes min, max, mean, std, percent-nonzero for each of the 1,080 process rate slots. Returns a list of dicts keyed by (var_name, slot_number). Severity is always INFO — this is a data exploration check.

Implementation notes: KEY_VARS subset, all 30 slots per variable

Check 2: Derivative Consistency

`check_2_derivative_consistency(time, rates, sv_time, sv_concs, sv_names, box_id)`

For each key state variable: (a) sums signed process rates to get kinetic dC/dt , (b) computes numerical dC/dt from $\Delta C/\Delta t$, (c) computes Pearson correlation, RMSE, (d) decomposes into %kinetic vs %transport contribution. A low correlation indicates transport dominance (expected for well-mixed variables like INORG_C).

Implementation notes: KEY_VARS, correlation > 0.1 threshold for inclusion

Check 3: Cross-Variable Consistency

`check_3_cross_variable_consistency(time, rates, box_id)`

Fifteen+ sub-checks validating inter-variable relationships: (a) O_2 reaeration sign and magnitude, (b) DIA/CYN/OPA/FIX_CYN/NOST DOC excretion = sum of individual phyto excretions, (c) NH_4 production from respiration $\approx$ growth $\times$ N:C, (d) ZOO feeding = sum of prey grazing rates, (e) DET production = sum of death + non-grazed feeding, and more. Uses both absolute tolerances and relative percentage thresholds.

Implementation notes: Tolerances: DOC excretion $\max(1e-5, 1e-4 \times \max_total)$; checks may be INACTIVE when underlying rates are zero

Check 4: Bug-Fix Verification

`check_4_bug_fix_verification(time, rates, box_id)`

Verifies that previously-identified bugs remain fixed: (a) FIX_CYN O_2 production (slot 19) is non-zero when FIX_CYN is active, (b) N-fixation rates for FIX_CYN and NOST are populated.

Implementation notes: Specific slots hardcoded from FORTRAN audit

Check 5: Dominant Processes

`check_5_dominant_processes(time, rates, box_id)`

For each key variable, ranks all contributing process rate slots by mean absolute magnitude and reports the top 3 slots with their percentage contribution to total flux. Helps identify which biogeochemical pathways drive each variable.

Implementation notes: Uses DERIVATIVE_SIGNS to select only slots that contribute to dC/dt

Check 6: Seasonal Patterns

`check_6_seasonal_patterns(time, rates, box_id)`

Computes seasonal means (DJF/MAM/JJA/SON) for key process rates (DIA growth, DIA respiration, ZOO feeding, nitrification, O_2 reaeration). Uses day-of-year from the time column (assuming 365-day years).

Implementation notes: Reports are INFO level; no automated anomaly detection

Check 7: Zero-Slot Analysis

`check_7_zero_slot_analysis(rates, box_id)`

Scans all NSTATE×NDIAGVAR slots for all-zero columns. Flags slots that (a) contribute to the derivative formula AND (b) are not expected to be zero given the model configuration (e.g., ZOO grazing on CYN being zero suggests low prey availability). Expected-zero slots (disabled features, AUX diagnostics) are marked as such.

Implementation notes: Uses SLOT_MAP derivative_slots_used to identify derivative-contributing slots

Check 8: Limitation Factors

check_8_limitation_factors(rates, box_id)

Extracts phytoplankton limitation factor slots (light, N, P, temperature, Si) from process rate AUX slots and reports their time-averaged min/mean/max. Values near 0 = strong limitation, 1.0 = no limitation. Covers DIA, CYN, OPA, FIX_CYN, NOST groups.

Implementation notes: Slot locations are hardcoded from SLOT_MAP AUX entries

Check 9: Negative Concentrations

check_9_negative_concentrations(sv_time, sv_concs, sv_names, box_id)

Checks all variables in NON_NEGATIVE_VARS for any negative values. Reports count, percentage, minimum value, and the timestep of first occurrence. Also checks dissolved oxygen saturation upper bound.

Implementation notes: ERROR if any negatives; WARNING if super-saturation detected

Check 10: NaN/Inf Detection

check_10_nan_inf(sv_concs, sv_names, pr_rates, box_id)

Scans all state variable columns and the bulk process rate matrix for NaN and ±Inf values. Reports per-variable counts.

Implementation notes: ERROR severity for any detection

Check 11: Stoichiometric Ratios

check_11_stoichiometry(sv_concs, sv_names, pr_rates, box_id)

Validates organism and detritus C:N:P ratios: ZOO N:C (expected ~0.22), ZOO P:C (~0.024), DET N:C, DET P:C, DOM N:C, DOM P:C. Also checks DIA growth O₂:C production ratio and ZOO respiratory N:C ratio. Uses median (robust to outliers) for main ratios.

Implementation notes: Thresholds: ±50% for ZOO, INFO for dynamic DET/DOM ratios. DET P:C classified as INFO (expected to be ■ Redfield)

Check 12: Sign Violations

check_12_sign_violations(pr_rates, box_id)

Checks all process rate slots listed in NONNEG_RATE_SLOTS for any negative values. This includes phytoplankton growth, respiration, excretion, death, ZOO feeding, akinete dynamics, and nitrification. Reports count, minimum value, and affected timesteps.

Implementation notes: WARNING if any negatives detected (previously found 18 violations from negative light limitation at dusk; now resolved)

Check 13: Euler Integration

check_13_euler_integration(sv_time, sv_concs, sv_names, pr_rates, box_id)

Reconstructs ΔC using forward Euler from kinetic dC/dt and compares to actual ΔC. The difference = transport flux. Reports %kinetic, %transport, correlation, and mean signed transport residual for each key variable.

Implementation notes: INFO level; transport-dominated variables (INORG_C, TOT_ALK) expected to have low %kinetic

Check 14: Allelopathy

check_14_allelopathy(sv_concs, sv_names, pr_rates, box_id)

Checks secondary metabolite state variables (SEC_METAB_*) for: (a) always-zero concentrations (suggests feature may be disabled), (b) negative values, (c) correlation with parent phytoplankton biomass. Also reports process rate magnitudes for SEC_METAB variables.

Implementation notes: Current config: all allelopathy SEC_METAB slots are zero (feature inactive)

Check 15: Smoothness / Spikes

```
check_15_smoothness(sv_time, sv_concs, sv_names, box_id)
```

For each key variable, computes day-to-day absolute changes $|\Delta C|$. A "spike" is a timestep where $|\Delta C| > \text{threshold}$, where $\text{threshold} = \max(10 \times \text{mean}|\Delta C|, \text{min_spike_size})$ and $\text{min_spike_size} = \max(1e-10, 0.001 \times \text{dynamic_range})$. Severity: WARNING if $n_spikes > 10$ AND spike magnitude $> 1\%$ of mean concentration; otherwise INFO.

Implementation notes: Also detects monotonic trends ($>80\%$ same direction = persistent drift)

Check 16: Mass Balance

```
check_16_mass_balance(pr_rates, box_id)
```

Computes total kinetic dN/dt , dP/dt , $dOrgC/dt$, $d(AlC)/dt$, and dO_2/dt by summing all process rate contributions across all state variables for each element. A perfectly closed model would give zero total kinetic flux. Non-zero values indicate external exchange (atmosphere, sediment, rivers) or numerical error.

Implementation notes: Uses default stoichiometric ratios to convert phytoplankton C to N/P equivalents

5. Shared Utilities (aquabc_analysis_utils.py)

Constants:

NDIAGVAR	30	Number of diagnostic/process rate slots per variable
NSTATE	32	Core state variables
NUM_ALLELOPATHY	4	Allelopathy secondary metabolite variables
NSTATE_TOTAL	36	Total state variables (32 + 4)
NUM_PROCESS_RATES	1080	36 × 30 process rate slots per box
BOX_IDS	[5,6,8,9,14,17,25]	Boxes producing output
BOX_TYPES	{5:sand, ..., 25:mud}	Sand vs mud classification
DEFAULT_N_TO_C	0.220	Default N:C ratio (mg N / mg C)
DEFAULT_P_TO_C	0.024	Default P:C ratio (mg P / mg C)
DEFAULT_O2_TO_C	2.66	Default O ₂ :C ratio
DEFAULT_Si_TO_C	0.25	Default Si:C ratio (diatoms)

Functions:

get_slot_col(var_idx, slot)	Returns 0-based column index in the process rate array
load_process_rates(dir, box)	Loads PROCESS_RATES.out file; returns (time, rates_2d)
load_state_vars(dir, box)	Loads PELAGIC_BOX.out file; returns (time, concs, header_names)
compute_kinetic_deriv(rates, var, slot_map, signs)	Sums signed process rates for dC/dt
compute_numerical_derivative(time, conc)	Forward differences $\Delta C / \Delta t$
find_sv_column(var_name, sv_names)	Case-insensitive column lookup in state var headers

6. Process Rate Slot Map (process_rate_slot_map.py)

This file contains the complete mapping from AQUABC Fortran source code to Python analysis. It defines two main data structures:

SLOT_MAP dictionary:

Keys are state variable names (e.g., "NH4_N", "DIA_C"). Each entry contains: var_index (1-based Fortran index), slots (dict mapping slot number → human-readable description), derivative_formula (string like "+1 +2 -3 -4"), derivative_slots_used (list of slot numbers), conditions (compile-time flags like DO_NOSTOCALES).

DERIVATIVE_SIGNS dictionary:

Keys are state variable names. Values are dicts mapping slot_number → sign (+1 or -1). Used by compute_kinetic_deriv() to reconstruct the DERIVATIVES array from PROCESS_RATES.

7. Usage Guide

Prerequisites:

- Python 3.8+ • numpy • reportlab (for PDF generation only)
- Model output files in OUTPUTS/ directory (must include PROCESS_RATES.out files — enable "Process Rates" output in Model Config → Output Config tab)

Command-line usage:

Run 16-check analysis (text output):

```
python tools/deep_process_rate_analysis.py --output-dir OUTPUTS
```

Generate analysis results PDF:

```
python tools/generate_analysis_results_pdf.py --output-dir OUTPUTS
```

Generate detailed analysis PDF:

```
python tools/generate_deep_pr_pdf.py --output-dir OUTPUTS
```

Run 10-check crosscheck:

```
python tools/deep_state_vs_process_crosscheck.py --output-dir OUTPUTS
```

Generate crosscheck PDF:

```
python tools/generate_crosscheck_pdf.py --output-dir OUTPUTS
```

Programmatic usage:

The analysis can be invoked programmatically from Python:

```
from tools.deep_process_rate_analysis import run_analysis
results = run_analysis("OUTPUTS")
# results is a dict: {box_id: {check_name: findings_list}}
# Each finding is a dict with "severity", "variable", and check-specific keys
```

8. Severity Classification

ERROR	Model-critical bug requiring code fix	NaN/Inf values, negative concentrations in variables that must be ≥ 0 , fundamental stoichiometric failures
WARNING	Ecological concern worth investigating	Significant smoothness spikes (>10 occurrences, >1% of mean concentration), unexpected zero slots for active processes, sign violations in process rates
INFO	Informational note, no action needed	Rate statistics, seasonal patterns, dominant processes, expected ratio deviations (e.g., DET P:C ■ Redfield), minor spikes in low-biomass species
OK	Check passed without issues	No negatives, no NaN/Inf, mass balance within tolerance

9. Extending the Analysis

To add a new check:

1. Create a function `check_N_name(args)` in `deep_process_rate_analysis.py` that returns a list of finding dicts, each containing at minimum a `severity` key.
2. Add the call in `run_analysis()` within the per-box loop.
3. Add reporting in `print_summary()`.
4. (Optional) Add a PDF section in `generate_deep_pr_pdf.py`.
5. Update the check count in the docstring and report titles.

To modify thresholds, edit the constants in each check function. Key thresholds:

Spike detection	$10 \times \text{mean}[\text{dC}]$ with floor of $0.001 \times \text{range}$	check_15
Spike severity	$n_{\text{spikes}} > 10$ AND magnitude $> 1\%$ of mean conc	check_15
DOC excretion tolerance	$\max(1e-5, 1e-4 \times \text{max_total})$	check_3
Mass balance imbalance	10% of mean individual variable flux	check_16
Stoichiometric match	$\pm 50\%$ for ZOO, INFO for DET/DOM	check_11